

MANAV DHAMECHA

GitHub | LinkedIn | Portfolio | manavdhamecha1@gmail.com | +919714110365 | Surat, India, 395009

Education

Sardar Vallabhbhai National Institute of Technology (SVNIT), Surat

2024 – Present

B.Tech in Artificial Intelligence

CGPA: **8.73** (Present)

Relevant Coursework: Data Structures, Design and Analysis of Algorithms, Operating Systems, Object Oriented Programming, Computer Architecture, Computer Networks, Database Management Systems, Discrete Mathematics.

Work Experience

Software Engineer Intern, Fastrac

May 2025 – July 2025

PHP, JavaScript, RESTful APIs, Google Maps

- Architected a WooCommerce shipping plugin integrating 3+ third-party logistics APIs, automating **100%** of shipment creation, tracking, cancellation, and label-printing workflows that were previously manual.
- Optimized checkout rate-calculation with API response caching and batch processing to reduce latency for high-volume stores.
- Built fault-tolerant backend workflows for bulk order processing, removing manual administrative effort.

Technical Skills

- **Languages:** C/C++, Python, TypeScript, JavaScript
- **Backend & Databases:** FastAPI, Flask, PostgreSQL, MySQL, Redis, REST APIs, gRPC
- **Systems & Tools:** Linux, Docker, Git, GitHub, CI/CD, AWS, CMake

Projects

Trading Hub - High-Performance Market Microstructure Analysis Platform

GitHub

C++17, gRPC, FastAPI, PostgreSQL, TimescaleDB, Docker, AWS

- Engineered a C++17 analytics engine integrated via gRPC, achieving **0.5ms** average feature-computation latency – a **4.4×** improvement over the Python implementation – computing **40+** real-time market microstructure features.
- Built a real-time ingestion and analytics pipeline processing **160+** order-book snapshots/sec with **6.9ms** end-to-end latency, from data ingestion through analytics to UI.
- Designed a time-series data layer on PostgreSQL + TimescaleDB storing **1.3M+** order-book snapshots at an **8:1** compression ratio, deployed via Docker on AWS to support live replay and anomaly detection.

Incognito Chat - Privacy-Focused Self-Destructive Chat Platform

GitHub Live

Next.js, TypeScript, Redis + Realtime (Upstash), Elysia.js

- Engineered a Redis-backed Pub/Sub messaging service using TTL-based self-destructing rooms, cutting persistent message storage from **100% to 0%**.
- Built real-time messaging with sub-**200ms** end-to-end latency per message.
- Reduced operational overhead via transient Redis storage and anonymous token-based auth, eliminating the need for a persistent user database.

Achievements

- **1st Place**, PowerMind Hackathon 2026 organized by **Adani Group & CDC, NIT Surat** (180+ participants)
- **1st Place**, Echelon 2026, a national 24-hour hackathon organized by **ISEA & ACM, NIT Surat** (45+ competing teams)

Publications

- Co-author, 2 publications in the **ACL Anthology** on multilingual NLP (AAACL-IJCNLP 2025, IIT Bombay)

Positions of Responsibility

• **Core Member**, ACM Student Chapter, NIT Surat

June 2026 – Present

• **Co-Head**, Mindbend, NIT Surat

Oct 2025 – Present